

- 4 (a) e.g. store energy (do not allow 'store charge')
 in smoothing circuits
 blocking d.c.
 in oscillators
any sensible suggestions, one each, max. 2

B2 [2]

- (b) (i) potential across each capacitor is the same and $Q = CV$

B1 [1]

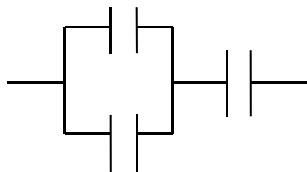
- (ii) total charge $Q = Q_1 + Q_2 + Q_3$
 $CV = C_1V + C_2V + C_3V$
 (allow $Q = CV$ here or in (i))
 so $C = C_1 + C_2 + C_3$

M1

M1

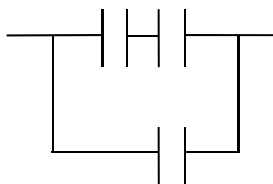
A0 [2]

- (c) (i)



A1 [1]

- (ii)



A1 [1]